Surya Kalimuthu

Kurukshetra, India | LinkedIn | GitHub

Education

National Institute of Technology, Kurukshetra CGPA: 7.48

B.Tech in Information Technology | 2023 – 2027 *Semester VI SGPA: 9.26*

Indian School Muscat 88.6%

CBSE Senior Secondary (Class XII) | 2023

Experience

FlyRank AI Jul. 2026 – Aug. 2026

Machine Learning Engineering Intern *Remote*

- Queried a 9.84M-row search-analytics dataset using DuckDB SQL over Parquet, checking for duplicate rows, date coverage, and missing analytics data.
- Trained Logistic Regression and Random Forest models to rank content pages for review, reaching 0.720 Precision@50 on unseen clients versus 0.520 for an impressions-only baseline.
- Showed that a random train-test split inflated Precision@50 to 0.940, versus 0.720 with clients held out, because the same clients appeared in both sets.

Projects

Distributed API Rate Limiter | *Node.js, Express.js, Redis, Lua, Jest, k6* GitHub

- Built a distributed token-bucket rate limiter for multiple Express instances using Redis-backed shared state and an atomic Lua script, preventing race conditions across concurrent requests.
- Implemented tier-based rate-limit policies, TTL-based cleanup of idle client buckets, and configurable fail-open/fail-closed behavior when Redis is unavailable.
- Validated concurrency and client isolation using 15 Jest integration tests and k6 load tests (300 virtual users, 9,900 requests), with zero cross-client throttling errors.

HealthLedger | *React, Express.js, MongoDB, Mongoose, JWT, Cloudinary* Live | GitHub

- Built a full-stack family health records platform with React and Express, supporting multiple profiles, medications, appointments, and medical document management.
- Implemented JWT-based authentication and profile-level access control to secure family health records, along with failure-safe Cloudinary workflows to prevent orphaned files and dangling database references.

Behavioral Drift Detection | *PySpark, Python, Pandas, NumPy, scikit-learn* GitHub

- Built a PySpark ETL pipeline to process 405K Google Borg cluster-trace records, retaining 349K records and engineering CPU and memory usage features.
- Analyzed resource-usage drift across 10 chronological windows using KS tests and PSI; trained a Random Forest failure classifier achieving 0.873 ROC-AUC and 0.816 average precision.

Technical Skills

Languages: Python, C++, SQL, JavaScript
Web: Node.js, Express.js, React.js, REST APIs, JWT Authentication, HTML, CSS
Databases: MongoDB, Redis
ML & Data: PySpark, Pandas, NumPy, scikit-learn, DuckDB, Parquet
Developer Tools: Git, GitHub, Jest, k6
Core CS: Data Structures & Algorithms (400+ LeetCode problems solved), OOP, DBMS, OS, Computer Networks

Positions of Responsibility

Social Media Head 2026 – Present

English Literary and Debating Club, NIT Kurukshetra

- Lead a 10+ member team as Social Media Head, organizing institute-level debates, quizzes, and Model United Nations events while coordinating publicity and event execution.